



International Islamic University Chittagong
Department of Computer Science and Engineering

LABORATORY MANUAL

Course Code: **CSE-1230**

Course Title: **Competitive Programming 1**

For the students of the 2nd Semester of B.Sc. in
Computer Science and Engineering program

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Course Objective

The main objectives of this course are

1. To introduce students to the algorithmic way of thinking and problem solving by computers.
2. To help the student to enhance their analyzing skills.
3. Foster creativity, teamwork, and innovation.

Preferred Programming Languages / Tools

- C
- C++

Preferred Text / Reference Books

SL	Name of Authors	Title of the book	Edition	Publisher's name	Year	ISBN
1	Steven Halim and Felix Halim	Competitive Programming 3	3rd	Lulu.com	2013	
2	Steven S Skiena, Miguel A. Revilla	Programming Challenges: The Programming Contest Training Manual	1st	Springer	2003	13: 978-0 38700 1630

Session 1

Title: Course Introduction and Polynomial Time Complexity

Topics covered:

- A roadmap of Competitive Programming up to the 4th semester [Video 1]
- CP1 syllabus, contests, evaluation, etc. [Video 1]
- Time complexity (Polynomial-time) [Video 2]

Recourses

- Video 1: S011 Intro to CP1
[<https://drive.google.com/file/d/1Nosz0lmMmvUnuw08UQoAiGKHiiXTRcGi/view?usp=sharing>]
- Video 2: S012 Time Complexity (Polynomial)
[<https://drive.google.com/file/d/1Vgqds3fdJkl3w3P85YAg-y1xLlKpEFzK/view?usp=sharing>]

Necessary Links:

- Tentative Course Outline:
<https://docs.google.com/spreadsheets/d/1ux-L44kerN1AEBbxshQFyS8UTRL91fARrmCkW-DpLYg/edit#gid=2091652929>
- Newcomer sheet: <https://codeforces.com/group/MWSDmqGsZm/contests>
- Junior Training Sheet:
https://docs.google.com/spreadsheets/d/1iJZWP2nS_OB3kCTjq8L6TrJJ4o-5lhxD0yTao cSYc-k/edit#gid=84654839

Practice Problems:

- 20 Problems from Newcomer Sheet #1
[<https://codeforces.com/group/MWSDmqGsZm/contest/219158>]
- 20 Problems from Newcomer Sheet #2
[<https://codeforces.com/group/MWSDmqGsZm/contest/219432>]

Session 2

Title: STL vector, string, iterator, and algorithm

Topics covered:

- STL vector, iterator, range-based loop, string [Video 1]
- STL algorithms - sort, reverse, find function [Video 1]
- Prefix sum

Recourses

- Video 1: S02 STL vector, string, iterator
https://drive.google.com/file/d/1UXHRBr8wKiPO-cWHyfSOT7ONgZppUO_a/view

Necessary Links:

- Leetcode string tagged problems:
https://leetcode.com/tag/string/?_cf_chl_captcha_tk__=pmd_ac4cd981f8b2c087cccfc54ee9d064b44ddd58e-1628072928-0-gqNtZGzNArijcnBszQyi
- Mostafa Saad STL script:
<https://raw.githubusercontent.com/mostafa-saad/ArabicCompetitiveProgramming/master/15%20C%2B%2B%20Programming%204%20Competitions/40%20C%2B%2B%20Programming%204%20Competitions%20-%20STL.cpp>
- Cpp reference STL: <https://www.cplusplus.com/reference/stl/>
- Geeks for geeks STL: <https://www.geeksforgeeks.org/the-c-standard-template-library-stl/>
- Topcoder STL (Part 1):
<http://community.topcoder.com/tc?module=Static&d1=tutorials&d2=standardTemplateLibrary>
- Topcoder STL (Part 2):
<http://community.topcoder.com/tc?module=Static&d1=tutorials&d2=standardTemplateLibrary2>

Practice Problems:

- 20 Problems from Newcomer Sheet #3 using STL vector
[<https://codeforces.com/group/MWSDmqGsZm/contest/219774>]
- 20 Problems from Newcomer Sheet #4 using STL string
[<https://codeforces.com/group/MWSDmqGsZm/contest/219856>]

Session 3

Title: Basic Math and Number Theory (Part 1)

Topics covered:

1. Prime checking in $O(\sqrt{N})$
2. Deriving mathematical equations
3. Basic of Bitwise operations
4. Concept of GCD & LCM
5. Prime factorization
6. Big Integer - addition, multiplication (reverse trick), division
7. Base conversion
8. Modular multiplication: $(A * B) \bmod M = (A \bmod M * B \bmod M) \bmod M$
9. Modular summation: $(A + B) \bmod M = (A \bmod M + B \bmod M) \bmod M$

Recourses

- Video 1: Sheet6_AtoQ
<https://drive.google.com/file/d/1JhByH9QKH0x1TqWwXwHZpz28dBHLKdPM/view>

Practice Problems:

- Problem A to Q from Newcomer Sheet #6
[<https://codeforces.com/group/MWSDmqGsZm/contest/223338>]

Session 4

Title: Basic Math and Number Theory (Part 2)

Topics covered:

[In video 1]

1. Precision error: double vs float
2. Problem R: intuition behind distance between points formula
3. Importance of drawing geometry-related problems
4. Problem S: Triangle-related formulas and their derivations
5. Meaning of determinant (A nice detailed playlist on linear algebra could be found here: <https://www.youtube.com/c/3blue1brown/playlists>)
5. Problem T: General formula of circle
6. Problem U: Solving line-related problems in different ways. Using slope and triangle area
7. Problem Y: Formulas for multidimensional space

[In video 2]

1. Solution of problem O
2. Problem E: A binary search approach to solve it

Recourses

- Video 1:
<https://drive.google.com/file/d/1jsOgFVhUjeNs2WlxiBEYNOP1SunDr1LM/view>
- Video 2:
<https://drive.google.com/file/d/1v0voPtmtqMmF1TT-UnyNnAXIKM8xR2-V/view>

Practice Problems:

- Problem R to Z from Newcomer Sheet #6
[<https://codeforces.com/group/MWSDmqGsZm/contest/223338>]
- Junior training sheet - A [JTS A - Row 11 to Row 22]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=1160016643>

Session 5

Title: Elementary Math for CP

Topics covered:

[In video 1]

1. Precision error: double vs float [Precision Error might become an issue if the $\log_2$ function is used: https://atcoder.jp/contests/abc215/tasks/abc215_b]
2. Data type representation in memory
3. Comparing double values
4. Math tricks for CP
5. Different library functions
6. Concept of log

Recourses

- Video 1:
<https://drive.google.com/file/d/1j0qM7HBxWAXLtu7N7qr9zaVQ-9HsMXVg/view>

Practice Problems:

- Junior training sheet - A [Row 23 to Row 44]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=1160016643>

Session 6

Title: Modular Arithmetic and Counting principles

Topics covered:

- [In video 1]
 1. Important chapters from Discrete Mathematics book - 4, 6, 8, 7
 2. Basics of Mod
 3. Some facts about mod
 4. Problem Discussion - UVA 408, UVA 10110
- [In video 2]
 1. Problem Discussion - UVA 12148, UVA 11231
- [In video 3]
 1. Discrete mathematics and its application - Chapter 6.1
 2. Sum rule and related problems
 3. Product rule and related problems
 - All possible combination printing using nested loops
 - All possible subset printing using bitmask
 4. Subtract rule and related problems (Inclusion - exclusion principle)
 - How many numbers between 1 and N are divisible by 2, 3, 5, or 7? (using bitmask)
 5. Division rule and related problem
 6. Approaching a problem in various ways
- All possible permutation and related problem

Recourses

- Video 1:
<https://drive.google.com/file/d/10pdnSmWrcMoxIifAIfQ5gHHMMwe1keqJ/view>
- Video 2:
https://drive.google.com/file/d/17J1SKyxgT2sL9dSdW_Qd9hR8uPTiOSNv/view
- Video 3:
https://drive.google.com/file/d/17J1SKyxgT2sL9dSdW_Qd9hR8uPTiOSNv/view
- All possible subset code:

How many numbers between 1 and N are divisible by N prime numbers such as 2, 3, 5, 7... (using bitmask)

<https://drive.google.com/file/d/1Qvn9hpOY68qNpqHc44maBf1bWbBFzoTj/view>

Practice Problems:

- All possible subset-related problem: https://atcoder.jp/contests/abc190/tasks/abc190_c
- Inclusion-exclusion: <https://www.spoj.com/problems/EASYMATH/>
- Junior training sheet - A [Row 49 to Row 88]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=1160016643>

By now students should solve at least 33 800-rated problems and 10 900-rated problems from codeforces.com.

During the midterm examination period and break afterward, students should complete 5 - 10 more 800-rated problems and 25 - 30 more 900-rated problems so that they feel comfortable with these ratings.

Session 7

Title: Geometry from Junior Training Sheet

Topics covered:

[In video 1]

1. Angle-related concepts and issues in CPP library
2. Basic trigonometry for CP
3. Area-related formulas for CP

[In video 2]

1. Discussion on line and circle
2. Different coordinate systems
3. Vector geometry for CP

Recourses

- Video 1:
https://drive.google.com/file/d/1cmLE2-x8_KxzVNxg-PvXthmq4bhBdSFv/view
- Video 2:
<https://drive.google.com/file/d/13VbcBF-114vQzB-MmFqNuByqct06wjLJ/view>

Practice Problems:

- Junior training sheet - A [Row 93 to Row 114]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=1160016643>

Session 8

Title: STL (Data Structure) Queue and Set

Topics covered:

[In video 1]

1. STL queue
 - Problems: Social networking easy version
 - Problems: Printing the first nonrepeating character of all prefixes of a string
2. STL - set
 - Problem - Social Networking - Hard version
3. **DS - BST and its time complexity**

Recourses

- Video 1: <https://drive.google.com/file/d/1IEOnzr3aIzGkZhacS2-IThrdeV7Jv9V-/view>

Practice Problems:

- Vjudge Long Contest [6 problems]
- Junior training sheet - B [Row 8 to Row 16]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=111761178>
- 10 more 1000 rated problems from codeforces.com

Session 9

Title: STL (Data Structure) Stack, Map, and Dequeue

Topics covered:

[In video 1]

1. Basics of stack
2. STL stack container
 - Balanced parenthesis problem
 - Finding the nearest larger element to the right in the list using stack.

[In video 2]

1. Basics of STL map
 - Building dictionary using STL map
 - Counting the number of elements in a list
2. Basics of STL dequeue
 - Related problem: <https://codeforces.com/problemset/problem/977/D>
3. STL multiset

Recourses

- Video 1:
https://drive.google.com/file/d/1_XRoTr_FVzaPASnj_OHawN-WPYhGCOa1/view
- Video 2:
<https://drive.google.com/file/d/1RNHRWlgzswLx3DEWkAUVfUFvxGsmLA6j/view>

Practice Problems:

- Vjudge Long Contest [6 problems]
- Junior training sheet - B [Row 28 to Row 36]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=111761178>
- 10 more 1000 rated problems from codeforces.com

Session 10

Title: STL sorting on Multiple fields

Topics covered:

[In video 1]

1. Basics of sorting on multiple fields
2. Comparator function as third parameter
3. Related problem discussion
 - UVA 10062: Tell me the frequencies
 - Finding the first student from a list based on different criteria

Recourses

- Video 1:
https://drive.google.com/file/d/1NtWHdGIMG_J_fKvLfNuphcU-9tPI02KL/view

Practice Problems:

- Vjudge Long Contest [6 problems]
- Junior training sheet - B [Row 47 to Row 55]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=111761178>
- 10 more **1100** rated problems from codeforces.com

Session 11

Title: Recursion

Topics covered:

[In video 1]

1. Basics of recursion
2. Recursion Tree
3. Printing from 1 to N, Fibonacci

[In video 2]

1. - Drawing recursion tree- Solving Problems using recursion
2. 5 steps to remember:
 1. Magical function
 2. Determining States
 3. Solving Sub problem
 4. Do necessary works for the current state
 5. Base case
3. Problem discussion: Printing Triangle from 1 to N, from N to 1, Digit Sum

[In video 3]

1. Solving array or string-related problems using recursion
2. Solving 2D array related problems using recursion
3. Problems: Array sum, string lowercase letter count, 2D array sum.

Recourses

- Video 1: https://drive.google.com/file/d/1yz7jkiWTd3_rlGEJvnpgG7nCJLQi-IsQ/view
- Video 2: <https://drive.google.com/file/d/1szDKy-nZKn2Yn8m77Pg5-8qRcOk2nC/view>
- Video 3: <https://drive.google.com/file/d/1jKtmNCvcpudCA3jYsINK-Pi-xu3ItW9c/view>

Practice Problems:

- 20 Problems from Neowcomer Sheet #7:
<https://codeforces.com/group/MWSDmqGsZm/contest/223339>
- 10 more **1100** rated problems from codeforces.com

Session 12

Title: Binary Search (Part 1)

Topics covered:

[In video 1]

1. Basics of Binary search
2. LogN complexity
3. Lower bound
4. Upper bound

Recourses

- Video 1:
<https://drive.google.com/file/d/1f-rbgILCWnH4-yIMxMVPE3aow6qgSoEL/view>
- Codeforces EDU (Step 1): <https://codeforces.com/edu/course/2/lesson/6>

Practice Problems:

- 3 problems from Codeforces EDU step 1: <https://codeforces.com/edu/course/2/lesson/6>
- Junior training sheet - A [Row 117]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=1160016643>
- Junior training sheet - B [Row 68 to Row 77]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=111761178>
- 5 more **1200** rated problems from codeforces.com

Session 13

Title: Binary Search (Part 2)

Topics covered:

[In video 1]

1. Binary search for the answer

[In video 2]

1. Binary search on fractions

Recourses

- Video 1:
<https://drive.google.com/file/d/1pcXrUr5uZmNC7jlwH4U5KOoXad8Lgvzw/view>
- Video 2: <https://drive.google.com/file/d/1XdJTLJrNh98qZOrz36aIvnWikbAdqDwa/view>
- Codeforces EDU (Step 2): <https://codeforces.com/edu/course/2/lesson/6>

Practice Problems:

- 6 problems from Codeforces EDU step 2: <https://codeforces.com/edu/course/2/lesson/6>
- Junior training sheet - A [Row 118 to Row 120]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=1160016643>
- Junior training sheet - B [Row 68 to Row 77]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=111761178>
- 5 more **1200** rated problems from codeforces.com

Session 14

Title: Number Theory (Part 1)

Topics covered:

1. Prime numbers checking in $O(\sqrt{N})$ complexity [video 1]
2. Sieve Method [video 2]
3. Prime factorization in $O(\sqrt{N})$ complexity [video 3]
4. Prime factorization in $O(\log(N))$ complexity [video 4]
5. Sieve related problem discussion

[<https://www.youtube.com/watch?v=zWLVZ6qZEXc&list=PL5DyztRVgtRWblnyEHtajWijQffrfwLz6&index=8>]

Recourses

- Video 1:
<https://www.youtube.com/watch?v=GXY6orMXU4c&list=PL5DyztRVgtRWblnyEHtajWijQffrfwLz6&index=2>
- Video 2:
<https://www.youtube.com/watch?v=7zYMjXqdchc&list=PL5DyztRVgtRWblnyEHtajWijQffrfwLz6&index=4>
- Video 3:
https://www.youtube.com/watch?v=UZJ2nO_rTPg&list=PL5DyztRVgtRWblnyEHtajWijQffrfwLz6&index=6
- Video 4:
<https://www.youtube.com/watch?v=lFUggck4agk&list=PL5DyztRVgtRWblnyEHtajWijQffrfwLz6&index=10>

Practice Problems:

- Problem 1: Primality Test [<https://www.codechef.com/problems/PRB01>]
- Problem 2: Finding the Kth prime [<https://www.spoj.com/problems/TDKPRIME/>]
- Problem 3: Micro and Prime Prime
[<https://www.hackerearth.com/practice/math/number-theory/primality-tests/practice-problems/algorithm/micro-and-prime-prime-1/>]
- Junior training sheet - B [Row 20 to Row 26, and Row 116]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=111761178>
- 10 more **1200** rated problems from codeforces.com

Session 15

Title: Number Theory (Part 2)

Topics covered:

1. Binary Exponentiation [video 1]
2. Euclidian algorithm for GCD and Basics of Modular Arithmetic [video 2]
3. Modular Arithmetic (Part 1) [video 3]
4. Modular Arithmetic (Part 2) [video 4]
5. Sieve related problem discussion

[<https://www.youtube.com/watch?v=zWLVZ6qZEXc&list=PL5DyztRVgtRWblnyEHtajWijQffrfwLz6&index=8>]

Recourses

- Video 1:
<https://www.youtube.com/watch?v=lsdwBlk9YGI&list=PL5DyztRVgtRWblnyEHtajWijQffrfwLz6&index=7>
- Video 2:
https://www.youtube.com/watch?v=0X_5V4CzddM&list=PL5DyztRVgtRWblnyEHtajWijQffrfwLz6&index=14
- Video 3:
<https://www.youtube.com/watch?v=FBhYOdP4gZU&list=PL5DyztRVgtRWblnyEHtajWijQffrfwLz6&index=16>
- Video 4:
<https://www.youtube.com/watch?v=Aaq8Y2lwKRU&list=PL5DyztRVgtRWblnyEHtajWijQffrfwLz6&index=17>

Practice Problems:

- Problem 1: GCD Queries [<https://www.codechef.com/problems/GCDQ>]
- Problem 2: Arpa's hard exam and Mehrdad's naive cheat
[<https://codeforces.com/problemset/problem/742/A>]
- Problem 3: Modular GCD [<https://www.codechef.com/problems/GCDMOD>]
- Junior training sheet - B [Row 20 to Row 26, and Row 116]
<https://docs.google.com/spreadsheets/d/1cXYS-yVmB8nNtUg1XMOsD4X20XZgaJLXSwPVs9teQeg/edit#gid=111761178>
- 10 more **1200** rated problems from codeforces.com